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Are assertive messages more effective in live marketing? Matching effects of anchor type and message assertiveness on purchase intention

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ABSTRACT

Artificial intelligence (AI) technologies have been extensively integrated into marketing practices, with virtual anchors emerging as novel tools. However, the existing research has rarely explored the optimal message strategies that different types of anchors should employ. By implementing two online scenario-based experiments and one laboratory experiment, this study investigates the matching effect between anchor type (virtual vs. human) and message assertiveness (assertive vs. non-assertive) on purchase intention, unveiling the distinct arousal mechanisms underlying these effects. The results indicate that the interaction between the anchor type and message assertiveness positively influences purchase intention, mediated by two emotional responses: excitement and relaxation. Specifically, virtual anchors that use assertive messages evoke higher levels of excitement among consumers, thereby increasing purchase intention. Conversely, human anchors that employ non-assertive messages induce greater relaxation, leading to increased purchase intention. These findings add to the literature on live marketing and AI-driven marketing communications. Additionally, they offer practical guidance for optimizing message strategies tailored to virtual and human anchors, thereby advancing the strategic design of live-streaming marketing campaigns.

1. Introduction

Live-streaming via e-commerce platforms has emerged as a novel digital marketing tool, garnering global attention. By June 2024, the number of e-commerce live-streaming users in China had reached 777 million (CNNIC, 2024). Artificial intelligence (AI) integration has further invigorated this sector. Virtual anchors—AI-generated agents capable of mimicking human behavior during live-streaming—stand out as promising innovation brought by AI (Gao et al., 2023). Virtual anchors not only match the transactional and interactive functions of human anchors, but also offer greater efficiency and cost-effectiveness (Gao et al., 2024). However, virtual anchors are not complete substitutes for their human counterparts (Liu et al., 2025). Thus, it is relevant and essential to understand optimal strategies for deploying human and virtual anchors in live-streaming.

The message strategy employed by an anchor is a critical determinant of live-streaming effectiveness. This study focuses on the role of assertive message strategies in live marketing. Assertive messages have become increasingly prevalent in marketing communications because they are believed to create an impression of urgency that consumers find difficult to resist (Kronrod et al., 2022). Phrases like “Buy now” and “Click the link” are frequently employed to encourage immediate purchases (Gu et al., 2023). Conversely, non-assertive messages, such as “Have a try” and “It’s worth getting,” encourage consumers to consider the products at their own pace (Kim et al., 2017). Although both strategies are prevalent in live-streaming, their relative effectiveness remains unclear. This study proposes that the effectiveness of assertive messages depends on the type of anchor that delivers them.

Previous research has explored scenarios where assertive messages yield more favorable consumer responses and identified various factors that influence their effectiveness (e.g., product type and recipient characteristics). For example, assertive messages are more persuasive for hedonic products (Kronrod et al., 2012b). They are also more effective for anxious consumers who exhibit higher cognitive fluency and hold positive attitudes toward assertive messages (Wang, 2022). However, most studies have overlooked the role of anchor type and other message source characteristics in assertive strategy effectiveness within live-streaming contexts. Different anchor types may not be

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equally suited for employing assertive messages.

Live-streaming is uniquely positioned to elicit strong emotional responses from consumers, which makes emotional arousal a key psychological mechanism influencing decision making (Wang et al., 2025). Rich audiovisual content allows live-streaming to evoke emotions, such as excitement and relaxation. For example, the unique live-streaming format and time-limited promotional language often generate excitement by offering novelty and intensity (Ni and Ueichi, 2024; Xue et al., 2020). Anchors who smile more and engage in personal or informal conversations can promote relaxation by creating a more intimate shopping experience that emphasizes viewer autonomy (Lin et al., 2021; Zhang et al., 2023). Different types of anchors using different message strategies may produce varying levels of excitement or relaxation. Both excitement and relaxation have been shown to positively influence consumer decision-making (Gorn et al., 1997).

Despite these insights, prior research has not explored the combined effects of anchor types and assertive message strategies on consumer behavior during live-streaming. Moreover, the role of source characteristics in shaping the effectiveness of message assertiveness remains unclear. In addition, the differences between emotional arousal triggered by human and virtual anchors have not yet been thoroughly explored. This study examines how assertive and non-assertive message strategies interact with anchor type (virtual vs. human) to influence purchase intention. Drawing on language expectancy theory (Burgoon et al., 2002), we propose that different anchors elicit distinct communication expectations. When the message aligns with these expectations, its persuasive impact is amplified (Baek et al., 2015). From an arousal perspective (Smith and Apter, 1975), we investigated the psychological mechanisms behind these matching effects.

By integrating language expectancy theory and arousal theory, this study contributes to the literature on AI-driven marketing communication, message assertiveness, and live marketing by offering theoretical advancements and practical guidance. Specifically, our findings provide actionable insights for designing message strategies tailored to virtual and human anchors, enabling live-streaming platforms and marketers to evoke positive emotions in consumers and effectively drive purchase intention.

2. Literature review and hypotheses

2.1. The matching effect of anchor type and message assertiveness

Assertive messages are characterized by straightforward and explicit instructions which reduce the message's ambiguity (Milfeld and Pittman, 2024). They enhance clarity of communication between the message recipient and provider. Conversely, non-assertive messages emphasize consumer autonomy by empowering them with the freedom to accept or reject the message (Kim et al., 2017). Table 1 summarizes existing studies on message assertiveness.

This study posits that the effectiveness of assertive message strategies in live-streaming varies according to the anchor type (virtual vs. human). According to language expectancy theory, the success of a message depends on whether its source aligns with the recipient's expectations (Burgoon et al., 2002). When these elements are in alignment, recipients are more likely to accept the message and exhibit positive responses such as increased purchase intention (Vazquez et al., 2020). Language expectancy theory emphasizes that persuasion effectiveness depends not only on the message contents and language but also on the sender's identity and the recipient's expectations of the sender (Burgoon et al., 2002). Language expectancy theory therefore offers a perspective that is particularly well-suited for exploring the interaction between anchor type and message assertiveness in live-streaming. This theoretical framework considers the characteristics of the persuader and their influence on viewers' decision-making.

Assertive messages, which are often perceived as more direct than non-assertive messages, are particularly effective when delivered via virtual anchors. Virtual anchors, as AI-driven agents, tend to be viewed as impersonal and systematic compared to human anchors. This leads consumers to expect rapid and straightforward communication from virtual anchors (Luo et al., 2019). Research indicates that consumers value efficiency and directness in their interactions with AI agents, particularly in a service context (Huang and Rust, 2021). Virtual anchors also possess consistent and stable characteristics such as fixed appearances and behaviors that remain unaffected by emotions or external pressures (Liu et al., 2025). This consistency allows virtual anchors to deliver assertive messages without the variability inherent to human anchors (Hu and Ma, 2023). Consequently, compared with human anchors, virtual anchors are more likely to meet consumers' expectations and positively influence their purchase intention when delivering assertive messages.

By contrast, non-assertive messages align more closely with the characteristics of human anchors than virtual anchors. Non-assertive messages create an environment that empowers consumers to make autonomous decisions and promotes proactive engagement (Nuttavuthisit, 2010). Human anchors excel at building relationships, maintaining positive social dynamics, and fostering emotional resonance. It is therefore crucial for human anchors to use non-assertive messages to avoid alienating viewers (Ribino, 2023). Messages of this nature foster sincerity and intimacy, enhancing trust and identification with the anchor. These factors immerse viewers in the live-streaming experience, ultimately increasing their purchase intention (Shi et al., 2023). Hence, non-assertive messages with human anchors are more likely to meet consumers' expectations and positively influence their purchase behavior than those with virtual anchors. Thus, we propose the following hypotheses.

H1. Anchor type and message assertiveness interact in their effects on purchase intention.

H1a. Assertive messages with virtual anchors have a stronger impact on purchase intention than with human anchors.

H1b. Non-assertive messages with human anchors have a stronger impact on purchase intention than with virtual anchors.

2.2. The mediating role of excitement and relaxation

The physiological arousal continuum, ranging from calmness to intense emotional states, has been refined in psychological models to emphasize the two distinct dimensions of arousal: excitement (ranging from boredom to excitement) and relaxation (ranging from tenseness to relaxation) (Smith and Apter, 1975). Research has consistently shown that excitement is more desirable than boredom and that relaxation is preferable to tenseness (Gorn et al., 1997). Excitement is characterized by high arousal and positive valence, whereas relaxation is characterized by low arousal and positive valence (Kensinger, 2004). Both excitement and relaxation levels can fluctuate in response to external stimuli such as words, sounds, and images (Cui et al., 2023). Message style and sources are particularly influential in shaping these emotional responses, especially within real-time social interactions, such as live-streaming (Feng et al., 2024; Kuperman et al., 2014).

Arousal theory states that novel, unique, and intense stimuli generally lead to higher arousal, especially in the excitement dimension (Eysenck, 2012). Previous research reinforced this notion by demonstrating that excitement is frequently associated with stimuli that are imaginative, novel, and dynamic (Noseworthy et al., 2014; Silvia, 2005). In live-streaming, the unique appearances of virtual anchors create a stronger sense of novelty. This may evoke a feeling of coolness and stimulate visual excitement (Gao et al., 2025). Empirical evidence suggests that virtual influencers are more likely to be perceived as exciting than their human counterparts (Zhang et al., 2025). Furthermore, arousal theory asserts that individuals actively seek stimuli to achieve an optimal level of arousal (Wirtz et al., 2000). An assertive message, which can instill a sense of affirmation and prompt action in consumers (Kronrod et al., 2022), is a more intense stimulus than a non-assertive message. Thus, assertive messages, closely aligned with the characteristics of

Table 1
Summary of studies reviewed in message assertiveness.

Authors (Year)	Independent Variables	Moderator variable	Dependent Variables	Methodology	Major Findings
Kronrod et al. (2012a)	Language (assertive vs. non-assertive)	Perceived Issue Importance	Compliance	Laboratory studies; Field experiment	Compliance with and expectations for more assertive language occur when the message is in line with the recipient's perceived importance of the issue in environmental campaigns.
Kronrod et al. (2012b)	Language (assertive vs. non-assertive)	Product type (Hedonic vs. Utilitarian)	Compliance	Experimental studies	More assertive messages can cause greater compliance in the context of hedonic consumption and lower compliance in utilitarian consumption contexts.
Baek et al. (2015)	Message assertiveness (assertive vs. non-assertive)	Effort investment	Recycling intentions	Experimental studies	Assertive messages cause individuals to show more favorable attitudes and behavioral intentions toward recycling when they invest significant effort in following the requests.
Kim et al. (2017)	Message assertiveness (assertive vs. non-assertive)	Cultures (South Koreans vs. Americans)	Persuasion	Experimental studies; Laboratory studies	Non-assertive messages are more persuasive for Americans, but South Koreans find both assertive and non-assertive messages to be equally persuasive.
Zemack-Rugar et al. (2017)	Ad Type (assertive vs. non-assertive)	Relationship type (Committed vs. Uncommitted)	Ad/Brand liking; Monetary allocations	Experimental studies	Assertive ads elicit more reactance from committed than uncommitted consumers.
Reavey et al. (2018)	Humanization (overt vs. subtle)	Assertive language	Likelihood to donate; Brand and ad attitude; Purchase intention	Experimental studies	Overt humanization leads to more positive advertising outcomes when the ad uses assertive language.
Villarroel Ordenes et al. (2019)	Speech acts (assertive, expressive, directive); Figures of speech (alliteration, repetition); Consistency; Information action		Consumer message sharing	Text mining	Directive messages induce less consumer sharing than assertive messages. Assertive that feature alliteration triggers greater consumer sharing on social media platforms.
Wang and Zhang (2020)	Message assertiveness (assertive vs. non-assertive) × Product type (want vs. should)	Consumer power	Purchase intent	Experimental studies; Field experiment	For high-power consumers, using more assertive language to promote want (should) products will lead to stronger (weaker) purchase intent.
Baek et al. (2021)	Emoji (present vs. absent)	Message assertiveness (assertive vs. non-assertive)	Sustainable intentions	Experimental studies	Participants responded more positively to the assertive message that included emojis but more negatively when emojis were removed.
Bang et al. (2021)	Message assertiveness (assertive vs. non-assertive)	Price discount (discount vs. no discount); Cultures (Koreans vs. Americans)	Compliance with a Prosocial Message	Experimental studies	American consumers generally dislike assertive messages but feel reciprocal obligations if marketers include price discounts, whereas South Korean consumers accept both assertive and nonassertive messages without resistance, and discounts have no effects on persuasion.
Kronrod et al. (2022)	Language (positive vs. negative)	Message assertiveness (assertive vs. non-assertive)	Healthful behavior	Experimental studies; Field experiment	Positive language increases compliance when expressed assertively, and negative communication is more effective when expressed non-assertively.
Choi et al. (2023)	Price discount (discount vs. no discount) × Message assertiveness (assertive vs. non-assertive)	Purchase motivation (hedonic vs. utilitarian)	Behavioral intentions	Laboratory studies	Discounts alleviate the guilt associated with hedonic consumption, resulting in a willingness to comply with assertive prosocial messages. Nonassertive and assertive messages are equally persuasive for utilitarian consumption, regardless of price discount.
Milfeld and Pittman (2024)	Message framing (altercast frame vs. no-frame)	Environmental concern; Market share (low vs. high)	Brand attitudes; behavioral intentions	Experimental studies	For consumers with higher levels of environmental concern, the altercast frame is more effective than a no-frame assertive message when originating from high market share versus low market share brands. However, this gap is attenuated for lower levels of environmental concern consumers.
Huang et al. (2023)	Language (assertive vs. non-assertive)	Temporal distance (distant vs. near)	Perceived review helpfulness; Attitudes toward the reviewed business.	Experimental studies	Reviews using assertive language are more persuasive among consumers whose travel time is in the distant future.
Yu and Liu (2024)	Language (assertive vs. non-assertive)	Food type (organic food vs. conventional food)	Purchase intention	Experimental study	Assertive (vs. non-assertive) language increases consumers' intention to buy organic food, whereas non-assertive (vs. assertive) language increases consumers' intention to buy conventional food.
Wang et al. (2024)	Customers' self-concept clarity (high vs. low) × Ads (assertive vs. non-assertive)	Brand perception (competence vs. warmth).	Advertising effectiveness	Experimental study	Assertive advertisements are more effective than non-assertive ones among consumers with high self-concept clarity.

virtual anchors, can consistently evoke a sense of excitement.

Additionally, arousal theory posits that high-arousal positive emotions are associated with brain activities that empower individuals to pursue goals and seek new pleasures (McManus et al., 2024). Virtual anchors are often designed as fictional characters with distinct personalities and traits. This leads viewers to develop expectations and desires regarding their behaviors (Wang et al., 2025). Assertive messages, characterized by their brevity and directness, can amplify the distinctiveness and dramatization of a virtual anchor's persona (Reavey et al., 2018). When virtual anchors use assertive and expressive messages, they increase their sense of uniqueness and stimulate stronger emotional reactions from viewers, which results in greater excitement (Noseworthy et al., 2014).

Conversely, audiences are generally more familiar with human anchors. They are therefore less likely to evoke high arousal levels. Although assertive communication can increase arousal by creating urgency and encouraging consumers to act quickly (Kim et al., 2017), assertive messages from human anchors do not consistently achieve optimal arousal levels. Furthermore, human anchors are often perceived as having self-serving intentions like financial gain. This can undermine trust and emotional engagement when human anchors deliver assertive messages (Garvey et al., 2023). When viewers question the anchor's motivations, this can hinder positive emotional arousal (Kircanski et al., 2018). Consequently, assertive messages from human anchors are less likely to elicit excitement.

According to arousal theory, individuals with high arousal tend to act impulsively and proactively (Petrolini and Viola, 2020). Empirical studies established a positive relationship between excitement and impulse buying. For example, heightened excitement can activate themes and associations linked to impulse buying, ultimately triggering this very behavior (Verplanken and Herabadi, 2001). Therefore, heightened excitement elicited by virtual anchors' use of assertive messages is likely to lead to increased purchase intention. Accordingly, we propose the following hypothesis.

H2. Compared to human anchors, assertive messages delivered by virtual anchors lead to a higher level of excitement in consumers, which in turn leads to stronger purchase intention.

Relaxation, a dimension of arousal that is distinct from excitement, can alleviate mental and muscular tension (Herman, 1985). It is commonly believed that AI induces a sense of rigidity in individuals, whereas human relaxation is perceived as more natural (De Freitas et al., 2023). Moreover, humans exhibit greater empathy in their emotional expressions, and the perception of empathy can relieve tension (Liu-Thompkins et al., 2022). In addition, if individuals are initially in a relaxed state, they may continue to seek verbal tones consistent with the corresponding emotions (Kim et al., 2010). Non-assertive messages are typically gentler and more approachable, creating a relaxed atmosphere (Mannell et al., 1988). Therefore, non-assertive messages with human anchors can consistently evoke a sense of relaxation.

Furthermore, according to arousal theory, low-arousal positive emotions are associated with brain activity in neural substrates involved in affiliation, attachment, and feelings of safety, both in connection with familiar stimuli and others (McManus et al., 2024). Non-assertive messages align closely with consumers' expectations for human-to-human communication (Liu-Thompkins et al., 2022). Through natural and open expressions, human anchors can evoke a sense of affinity and calmness (Shi et al., 2023). This makes non-assertive messages with human anchors particularly effective at inducing relaxation. This can lower viewers' psychological defenses and foster more relaxed and enjoyable experiences (Pham and Sun, 2020).

By contrast, the programmed nature of virtual anchors may limit their ability to create the same level of emotional connection with viewers. This makes it more difficult for viewers to fully relax (Gao et al., 2024). Additionally, emotional arousal can reduce the attentional resources available for information processing. In doing so, attention is focused on the arousing stimulus (Kensinger, 2004). Viewers often direct an inordinate amount of attention to these virtual anchors, resulting in a scarcity of cognitive resources (Zhou et al., 2023). However, nonassertive messages that emphasize autonomy and self-reflection inherently require a greater allocation of cognitive resources (Kronrod et al., 2012a). When virtual anchors convey these non-assertive messages, viewers may be inadvertently compelled to exert more cognitive effort, ultimately hindering relaxation (David et al., 2024).

Arousal theory describes both relaxation and arousal as positive emotional states (Kensinger, 2004). When consumers relax, their positive mood often encourages them to spend more time engaging with products and increases their likelihood of making purchasing decisions (Chebat and Michon, 2003). Previous research has demonstrated that relaxation benefits consumer decision-making as it can enhance the likelihood of making favorable choices (Pham and Sun, 2020). For example, relaxed consumers opt for healthier food options (Biswas et al., 2019). Therefore, non-assertive messages with human anchors are expected to generate greater relaxation and, in turn, increase purchase intention more effectively than virtual anchors. This leads to the following hypothesis.

H3. Compared to virtual anchors, non-assertive messages with human anchors lead to a higher level of relaxation in consumers. This, in turn, leads to stronger purchase intention.

These inferences lead to the research model shown in Fig. 1.

3. Methodology

We conducted two online scenario-based experiments and one laboratory experiment to test the hypotheses (Table 2). Three everyday consumer products frequently sold by anchors during live-streaming sessions were selected: a thermos cup, a headset, and sunglasses. These products incorporate both search and experience attributes, appealing to diverse audiences to avoid potential gender biases that could influence the results. To further enhance the validity of the experiments and exclude brand awareness effects, none of the selected products contained identifiable brand information. This ensured that participants' responses were primarily influenced by the anchor types and message strategies rather than by preconceived notions about specific brands.

The selected products were also selected as experimental stimuli in previous studies, establishing a strong foundation for their relevance to this study (e.g., Affonso and Janiszewski, 2023; Chang and Hung, 2018; Zhou et al., 2023). This methodological design facilitated a focused analysis of how human and virtual anchors impact consumer excitement, relaxation, and subsequent purchase intention through different message strategies, the visual conceptual framework shown in Fig. 2.

4. Study 1

4.1. Research design and procedure

Study 1 examined the interaction between anchor type and message assertiveness on purchase intention. Participants were recruited using Credamo's comprehensive online sample database (www.credamo.com). Random sampling ensured diverse demographic backgrounds and experiences were represented. A total of 600 questionnaires were collected, of which 500 (83.3%) were deemed valid. We excluded responses from individuals without live-streaming experience, as well as questionnaires with logically inconsistent answers and those completed in less than 1 min. The final sample (62.8% female; $M_{age} = 30.85$ years, *range*: 18–72 years) was retained for analysis. Appendix A provides the detailed sample characteristics of all the studies. This study adopts a 2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) between-subjects experimental design.

Participants received an overview of the live-streaming concept, including examples of various anchors, to ensure a comprehensive understanding of the medium. Following this introduction, respondents were asked whether they had recently watched content produced by any

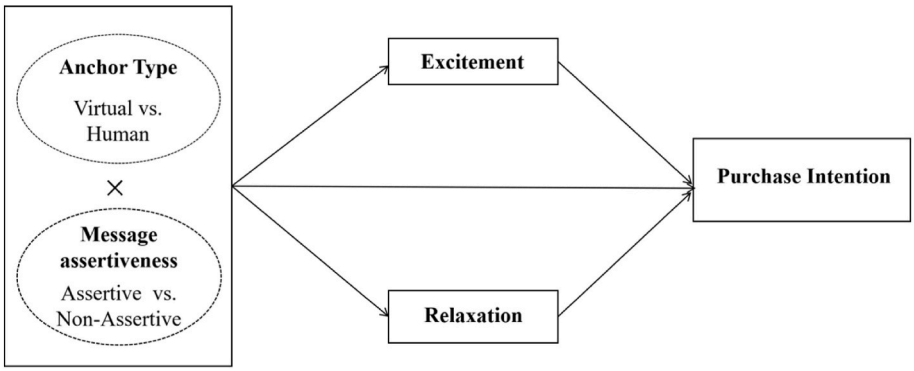


Fig. 1. Research model.

Table 2
Summary of studies.

Study	Method	Design	Hypothesis testing
Study 1	Online scenario-based experiments	2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) between-subjects design	H1a, H1b established
Study 2	Online scenario-based experiments	2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) between-subjects design	H1a, H1b, H2, and H3 established
Study 3	Laboratory experiment	2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) between-subjects design	H1a, H1b, H2, and H3 established

of the anchors described. If their responses were affirmative, participants were required to share the frequency of their viewership and the extent of their purchases from these anchors.

A series of targeted questions were asked related to participants' experiences with the specific anchors they had mentioned. These questions collected information on time spent watching, featured products, viewing platform, and participants' emotional experiences during the interaction. This approach aimed to enhance participants' recollections of live-streaming experiences and ensured that the analysis considered the context of their engagement (Voorveld et al., 2018).

Next, the participants were asked to imagine themselves as viewers of a live-streaming in which the anchor introduced a thermos cup as a featured product. We then manipulated message assertiveness using the method proposed by Baek et al. (2015), compiling two distinct test materials for message assertiveness. The assertive message was characterized by imperatives and controlling terms, such as "must," "should," and "ought." The non-assertive message emphasized the viewer's autonomous actions, incorporating terms like "could," "might want to," and "worth." In the assertive condition, the message conveyed was: 'You must buy it, no need to consider.' In the non-assertive condition, the message conveyed was: 'It's worth buying. You can consider it.' Apart from the manipulation of message assertiveness, all other factors such as product type, introduction, price, and product picture display remained the same (all experimental materials can be seen in Appendix B).

We then measured purchase intention ($\alpha = .92$) using the scale adapted from Aaker et al. (2010). The items were as follows: "I am very interested in purchasing the product introduced by the anchor; " "I am very likely to purchase the product introduced by the anchor; " and "I am very satisfied with the product introduced by the anchor." We also measured participants' perceptions of message assertiveness ($\alpha = .82$) using three items (Baek et al., 2015): "The message was forceful; " "The message was intense; " and "The message was powerful." All items were rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). Finally, the participants provided demographic information, including gender, age, educational background, and monthly income. Appendix C summarizes the measurements of the main variables, and Appendix D shows the descriptive statistics and correlations of the variables in all the studies.

4.2. Manipulation checks

An independent sample *t*-test indicated that assertive messages had a greater average score than non-assertive messages ($M_{assertive} = 5.69$ vs. $M_{non\text{-}assertive} = 4.06$; $t(498) = 16.24$, $p < .001$). This suggested that the experiment's manipulation of message assertiveness was successful.

4.3. Results

After controlling for all demographic variables, a 2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) Analysis of Variance (ANOVA) indicated that the main effects

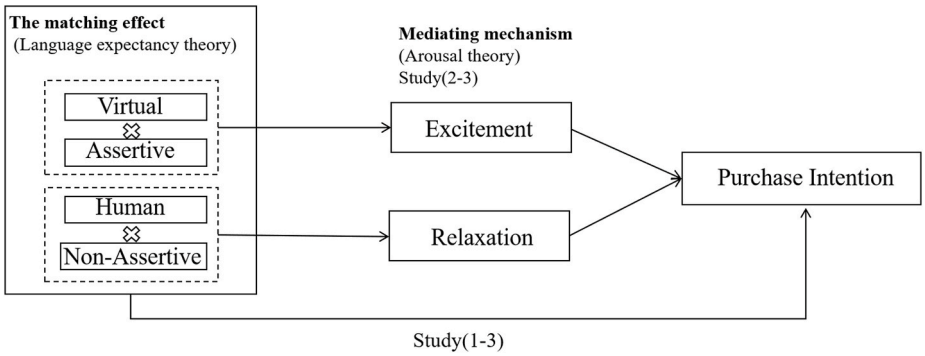


Fig. 2. Visual conceptual framework.

of anchor type ($F(1,492) = 1.08$, $p = .30$) and message assertiveness ($F(1,492) = 2.56$, $p = .11$) on purchase intention were not significant. However, the interaction between anchor type and message assertiveness was significant ($F(1,492) = 62.79$, $p < .001$, $\eta_p^2 = 0.11$). Fig. 3 and Table 3 show that, in subsequent comparisons, assertive messages with virtual anchors ($M = 5.54$, $SD = 1.04$) had a greater impact on purchase intention than with human anchors ($M = 4.47$, $SD = 1.39$; $F(1, 492) = 42.35$, $p < .001$, $\eta_p^2 = 0.08$). In contrast, non-assertive messages with human anchors ($M = 5.53$, $SD = 1.05$) had a greater impact on purchase intention than with virtual anchors ($M = 4.72$, $SD = 1.42$; $F(1, 492) = 22.86$, $p < .001$, $\eta_p^2 = 0.04$). These results support H1a and H1b.

4.4. Discussion

Study 1 demonstrates a matching effect between the anchor type and message assertiveness on purchase intention, thus supporting H1a and H1b. Consumer purchase intention was stronger when the virtual anchor used assertive messages and when the human anchor used non-assertive messages. Nevertheless, the mechanisms underlying this matching effect remain unknown. Study 2 was designed to investigate the mediating process in the association between the interaction (anchor type and message assertiveness) and purchase intention. Additionally, the study examined the robustness of the results observed in Study 1 using different experimental materials and new control variables.

5. Study 2

5.1. Research design and procedure

Study 2 aimed to replicate the results of Study 1 while examining the mediating mechanisms underlying the interaction between anchor type and message assertiveness on purchase intention. Questionnaires were distributed using Credamo's sampling platform, leveraging its proprietary algorithms to randomly target active online users. Of the 500 responses received, 431 (86.2%) were deemed valid after excluding illogical responses, questionnaires completed in less than 1 min, and questionnaires where respondents failed attention-check questions. The final sample (58.7% female; $M_{age} = 39.59$ years, *range*: 17–69 years) was retained for analysis. This study employed a 2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) between-subjects experimental design.

In Study 2, we generated images of both human and virtual anchors. These anchors were then inserted into the same live-streaming room. To control for other variables that may have affected the results, we conducted a pre-test ($N = 200$). conducted. The results identified no significant differences in the participants' perceived attractiveness,

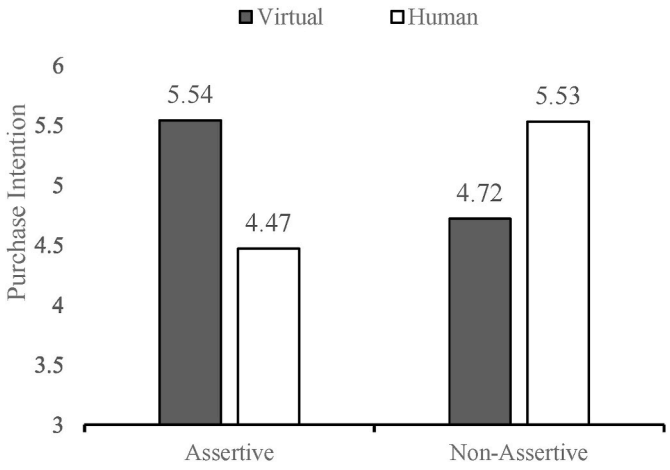


Fig. 3. Purchase intention as a function of anchor type and message assertiveness (Study 1).

Table 3
Summary of results on excitement, relaxation, and purchase intention.

Study	Factor	Excitement		Relaxation		Purchase intention		
		M	SD	M	SD	M	SD	N
Study 1	Virtual × Assertive	–	–	–	–	5.54	1.04	122
	Virtual × Non-Assertive	–	–	–	–	4.72	1.42	127
	Human × Assertive	–	–	–	–	4.47	1.39	130
	Human × Non-Assertive	–	–	–	–	5.53	1.05	121
Study 2	Virtual × Assertive	5.96	1.73	4.86	1.87	5.54	0.96	110
	Virtual × Non-Assertive	4.26	1.64	6.42	1.21	5.00	1.10	103
	Human × Assertive	5.38	1.62	5.09	1.14	4.85	1.40	109
	Human × Non-Assertive	4.36	1.75	6.84	1.03	5.48	0.93	109
Study 3	Virtual × Assertive	5.71	1.74	4.21	1.50	4.12	1.47	47
	Virtual × Non-Assertive	2.55	1.47	5.13	1.58	2.03	0.78	47
	Human × Assertive	3.32	2.03	3.90	1.71	2.47	1.10	47
	Human × Non-Assertive	2.87	1.39	5.86	1.73	4.00	1.49	47

credibility, expertise, familiarity, or general attitudes toward virtual and human anchors (see Appendix E for more details).

A headset was selected as the stimulus in Study 2. The participants were introduced to the live-streaming format, after which they were asked to recall whether they had ever watched live-streaming content. If the response was yes, the participants were shown either a virtual or a human anchor. They were then asked to identify the anchor and to report their frequencies of watching and shopping from the anchor type. The participants were then instructed to imagine that they were watching a live-streaming featuring this anchor, in which the anchor introduced a headset as a featured product. Study 2 used the same manipulation of message assertiveness as in Study 1.

We used a single-choice question to determine whether the participants watched a virtual or a human anchor. We then used the test method from Study 1 to examine the manipulation of message assertiveness ($\alpha = .85$). Referring to Gorn et al. (1997), we used three items ("relaxed," "soothed," and "calm") to measure relaxation ($\alpha = .86$) and two items ("excited" and "stimulated") to measure excitement ($\alpha = .76$). Next, we measured purchase intention ($\alpha = .89$) using the scale from Study 1, assessed participants' perceptions of product importance ($\alpha = .93$) using the scale adapted from Mittal (1989), and asked participants about their perceived price of the headset (Chang and Wildt, 1994). Finally, participants provided their demographic information.

We also measured several potential alternative explanations for the relationship under investigation, including information persuasion ($\alpha = .91$) (Keller and Block, 1996), perceived competence ($\alpha = .86$), and perceived warmth ($\alpha = .90$) (Wang et al., 2017).

5.2. Manipulation checks

All participants passed the anchor identification question manipulation test. An independent sample *t*-test indicated that assertive messages had a greater average score than non-assertive messages ($M_{assertive} = 5.73$ vs. $M_{non\text{-}assertive} = 4.25$; $t(429) = 13.59$, $p < .001$). Therefore, the experiment's manipulation of message assertiveness was successful.

5.3. Results

The anchor type (virtual = 0, human = 1) and message assertiveness (assertive = 0, non-assertive = 1) were dummy coded. We included

product importance, perceived price, and gender as control variables. A 2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) ANOVA indicated that the main effects of anchor type ($F(1, 424) = 0.24$, $p = .626$) and message assertiveness ($F(1, 424) = 0.80$, $p = .372$) on purchase intention were not significant. However, the interaction between anchor type and message assertiveness was significant ($F(1, 424) = 12.94$, $p < .001$, $\eta_p^2 = 0.03$). Fig. 4 shows, in subsequent comparisons, show that assertive messages with virtual anchors ($M = 5.54$, $SD = 0.96$) had a greater impact on purchase intention than with human anchors ($M = 4.85$, $SD = 1.40$; $F(1, 424) = 8.62$, $p < .01$, $\eta_p^2 = 0.02$). In contrast, non-assertive messages with human anchors ($M = 5.48$, $SD = 0.93$) had a greater impact on purchase intention than with virtual anchors ($M = 5.00$, $SD = 1.10$; $F(1, 424) = 4.83$, $p < .05$, $\eta_p^2 = 0.01$). These results replicated those obtained in Study 1.

The two-way ANOVA indicated that assertive messages with virtual anchors ($M = 5.96$, $SD = 1.73$) evoked higher levels of excitement than by human anchors ($M = 5.38$, $SD = 1.62$; $F(1, 424) = 5.89$, $p < .05$, $\eta_p^2 = 0.01$; see Fig. 5). In contrast, non-assertive messages with human anchors ($M = 6.84$, $SD = 1.03$) evoked higher levels of relaxation than with virtual anchors ($M = 6.42$, $SD = 1.21$; $F(1, 424) = 4.55$, $p < .05$, $\eta_p^2 = 0.01$; see Fig. 6). Non-assertive messages to evoke excitement ($F(1, 424) = 0.01$, $p = .943$) and assertive messages to evoke relaxation ($F(1, 424) = 2.98$, $p = .412$) did not differ between the virtual and human anchors.

Subsequently, the mediating effects of excitement and relaxation were tested using PROCESS Model 8. A bias-corrected bootstrap procedure was employed to generate 5000 resamples with 95% confidence intervals (*CI*). As shown in Table 4, when assertive messages were used, the mediating effect of excitement was significant (*effect* = -0.20, *95% CI* = [-0.35, -0.06]), but the mediating effect of relaxation was not (*effect* = -0.08, *95% CI* = [-0.20, 0.02]). Under this condition, assertive messages with virtual anchors increases purchase intention through excitement more than with human anchors. When using non-assertive messages, the mediating effect of excitement was not significant (*effect* = 0.07, *95% CI* = [-0.06, 0.21]), but the mediating effect of relaxation was significant (*effect* = 0.09, *95% CI* = [0.02, 0.17]). Under this condition, non-assertive messages with human anchors increase purchase intention through relaxation more than with virtual anchors. Therefore, H2 and H3 are supported.

Finally, we tested the alternative mechanisms separately using Model 8. The results showed that the mediating effects of perceived warmth (*effect* = 0.02, *95% CI* = [-0.06, 0.12]), perceived competence (*effect* = 0.02, *95% CI* = [-0.08, 0.31]), and perceived information persuasion (*effect* = 0.05, *95% CI* = [-0.15, 0.27]) were not significant in the interaction. Therefore, these variables were excluded from analysis as alternative mechanisms.

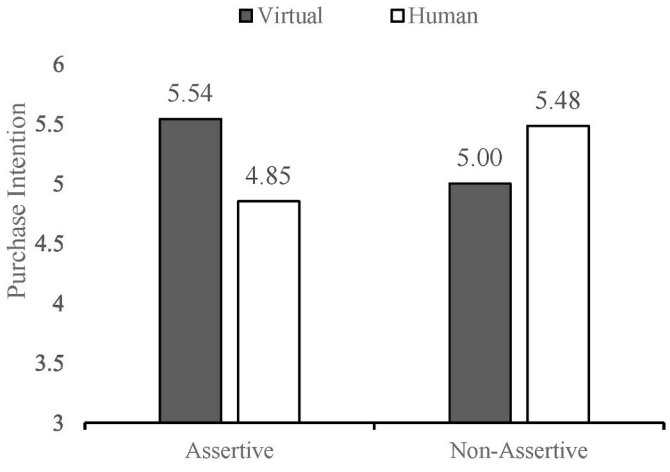


Fig. 4. Purchase intention as a function of anchor type and message assertiveness (Study 2).

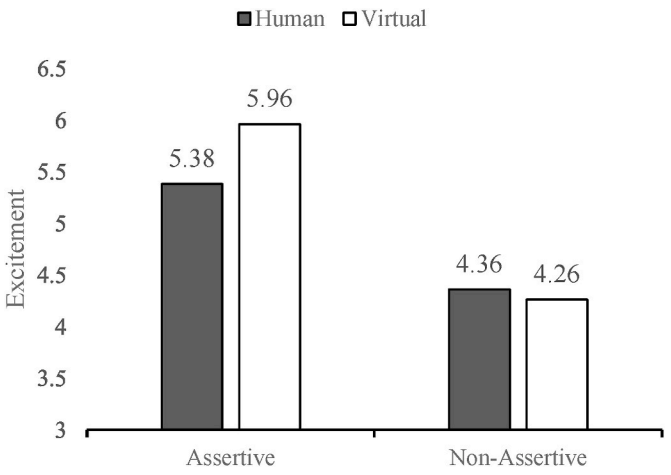


Fig. 5. Excitement as a function of anchor type and message assertiveness (Study 2).

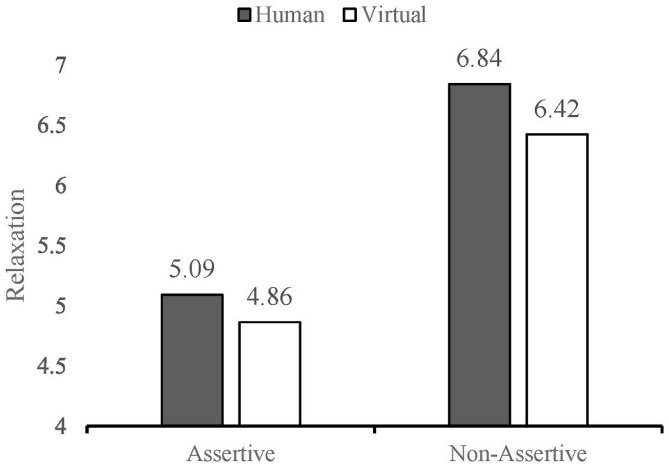


Fig. 6. Relaxation as a function of anchor type and message assertiveness (Study 2).

5.4. Discussion

Study 2 confirms the findings from Study 1 by demonstrating the matching effect of anchor type and message assertiveness on purchase intention. It also identifies the mediating effects of excitement and relaxation. The findings from Study 2 demonstrate that consumers may experience a higher level of excitement when exposed to assertive messages with virtual anchors and a higher level of relaxation when exposed to non-assertive messages with human anchors. These heightened levels of excitement and relaxation were associated with increased purchase intention. Three alternative explanations for the investigated relationship (perceived warmth, competence, and information persuasion) were excluded.

6. Study 3

6.1. Research design and procedure

Study 3 aimed to validate the robustness of the matching effect observed in Study 1 and Study 2. Additionally, Study 3 further explores the mediating mechanisms with a more realistic live-streaming scenario in a laboratory setting. The study recruited a convenience sample of 188 college students (72.3% female, $M_{age} = 21.95$ years) from diverse academic and professional backgrounds at a university in Changsha, China.

Table 4
Results of mediation analysis.

Study	Mediation	Path	Condition	*B*	*SE*	*LLCI*	*ULCI*
Study 2	Excitement	Direct effect	*M* - *SD* (Assertive)	−0.36	0.13	−0.63	−0.10
	Relaxation		*M* + *SD* (Non-Assertive)	0.36	0.14	0.10	0.63
		Indirect effect	*M* - *SD* (Assertive)	−0.20	0.07	−0.35	−0.06
			M + *SD* (Non-Assertive)	0.07	0.07	−0.06	0.21
		Indirect effect	*M* - *SD* (Assertive)	−0.08	0.06	−0.20	0.02
			M + *SD* (Non-Assertive)	0.09	0.04	0.02	0.17
Study 3	Excitement	Direct effect	*M* - *SD* (Assertive)	−0.52	0.16	−0.84	−0.19
	Relaxation		*M* + *SD* (Non-Assertive)	0.98	0.15	0.69	1.28
		Indirect effect	*M* - *SD* (Assertive)	−0.49	0.13	−0.77	−0.25
			M + *SD* (Non-Assertive)	0.11	0.07	−0.02	0.24
		Indirect effect	*M* - *SD* (Assertive)	−0.04	0.05	−0.14	0.05
			M + *SD* (Non-Assertive)	0.10	0.05	0.01	0.20

All participants had prior experience of watching live-streaming. This study employed a 2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) between-subjects design.

A student sample was selected for several reasons. According to a recent statistical report (CNNIC, 2024), college students are among the most active Internet users. Hence, they represent a key demographic for examining online behaviors, particularly in the context of live marketing (Zheng et al., 2023). Second, the experimental product, sunglasses, was aligned with college students' consumption preferences. Fashion items like sunglasses, are particularly relevant to this demographic, as these items are often used for self-expression (Djafarova and Bowes, 2021). Third, as Calder et al. (1981) suggest, a homogeneous sample can minimize confounding variables, thereby enhancing the internal validity and robustness of theoretical testing. Finally, many prior studies on live-streaming utilized student samples, demonstrating their appropriateness for this type of research (e.g., Tang et al., 2024; Wang, 2024; Yan et al., 2024; Yang and Ha, 2021; Zhang et al., 2020).

The experiments were conducted in a controlled laboratory setting. We created different conditions using live-streaming videos. To control for potential confounding variables, a pre-test ($N = 60$) was conducted in the laboratory setting. The results identified no significant differences in perceived attractiveness, credibility, expertise, familiarity, or general attitude toward virtual and human anchors across the four conditions (see Appendix F for more details).

In Study 3, sunglasses were selected as the stimulus. The manipulation of message assertiveness was consistent with Study 1. Participants were first asked to alphabetize a set of 10 words to control for potential mood differences caused by external factors beyond the laboratory setting (Wegener et al., 1995). Participants were then seated in individual booths to prevent visual contact. Each booth contained a computer, a pair of headphones, a description of the experiment, and a questionnaire. Participants were asked to look at product images of the sunglasses and then click on the 'Live Explanation' button in the top right-hand corner of the computer screen to watch a 46-s live video. Apart from the manipulation of the anchor and message assertiveness, all other aspects of the live video, including background, sound, and body movements, remained the same.

After watching the video, participants filled out paper questionnaires reporting their purchase intention ($\alpha = .93$), product importance ($\alpha = .93$), relaxation ($\alpha = .82$), and excitement ($\alpha = .81$). We then assessed their perceptions of message assertiveness ($\alpha = .80$). We also asked participants directly about willingness to pay (WTP): "What is the maximum you would pay for this product?" Compared with indirect methods, direct methods provide a more accurate estimate of WTP (Schmidt and Bijmolt, 2020). Finally, participants provided their demographic information. We also measured a few other variables, including positive emotion ($\alpha = .84$) (Kronrod et al., 2012b) and perceptive enjoyment ($\alpha = .90$) (Soares and Pinho, 2014).

6.2. Manipulation checks

All participants passed the anchor identification question manipulation test. In addition, the independent samples *t*-test indicated that the average score for assertive messages was greater than that for non-assertive messages ($M_{assertive} = 4.53$ vs. $M_{non\text{-}assertive} = 2.657$; $t(186) = 12.445$, $p < .001$). Therefore, the manipulation was successful.

6.3. Results

We included product importance, monthly living expenses, and gender as control variables. A 2 (anchor type: virtual vs. human) × 2 (message assertiveness: assertive vs. non-assertive) ANOVA indicated that the main effects of anchor type ($F(1, 181) = 0.38$, $p = .564$) and message assertiveness ($F(1, 181) = 2.30$, $p = .131$) on purchase intention were not significant. However, the interaction between anchor type and message assertiveness was significant ($F(1, 181) = 90.20$, $p < .001$, $\eta_p^2 = 0.33$). Fig. 7 shows, in subsequent comparisons, assertive messages with virtual anchors ($M = 4.12$, $SD = 1.47$) had a greater impact on purchase intention than with human anchors ($M = 2.47$, $SD = 1.10$; $F(1, 181) = 39.98$, $p < .001$, $\eta_p^2 = 0.18$). In contrast, non-assertive messages with human anchors ($M = 4.00$, $SD = 1.49$) had a greater impact on purchase intention than with virtual anchors ($M = 2.03$, $SD = 0.78$; $F(1, 181) = 51.12$, $p < .001$, $\eta_p^2 = 0.22$). These results replicated the results of Study 1 and Study 2.

Furthermore, the two-way ANOVA indicated that the main effects of anchor type ($F(1, 181) = 2.00$, $p = 0.159$) and message assertiveness ($F(1, 181) = 0.49$, $p = .487$) on WTP were not significant. However, the interaction between anchor type and message assertiveness was significant ($F(1, 181) = 13.10$, $p < .001$, $\eta_p^2 = 0.07$). Subsequent comparisons

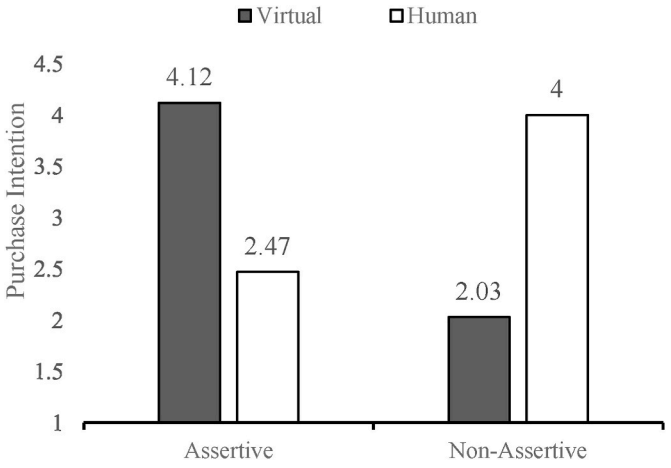


Fig. 7. Purchase intention as a Function of anchor type and Message assertiveness (Study 3).

showed that assertive messages with virtual anchors ($M = 130.06$ CNY, $SD = 98.10$) could lead to higher WTP than with human anchors ($M = 104.40$ CNY, $SD = 73.56$; $F(1, 181) = 9.41$, $p < .005$, $\eta_p^2 = 0.05$). In contrast, non-assertive messages with human anchors ($M = 142.51$ CNY, $SD = 129.62$) could lead to higher WTP than with virtual anchors ($M = 65.47$ CNY, $SD = 56.17$; $F(1, 181) = 4.33$, $p < .05$, $\eta_p^2 = 0.02$). These results support H1a and H1b.

The two-way ANOVA also indicated that assertive messages with virtual anchors ($M = 5.71$, $SD = 1.74$) evoked a higher level of excitement than with human anchors ($M = 3.32$, $SD = 2.03$; $F(1, 181) = 48.74$, $p < .001$, $\eta_p^2 = 0.21$). In contrast, non-assertive messages with human anchors ($M = 5.86$, $SD = 1.73$) evoked a higher level of relaxation than with virtual anchors ($M = 5.13$, $SD = 1.58$; $F(1, 181) = 4.83$, $p < .05$, $\eta_p^2 = 0.03$). Non-assertive messages to evoke excitement ($F(1, 181) = 2.22$, $p = .138$) and assertive messages to evoke relaxation ($F(1, 181) = 0.69$, $p = .406$) did not differ between the virtual and human anchors.

Subsequently, the mediating effects of excitement and relaxation were tested using PROCESS Model 8. As shown in Table 2, when assertive messages were used, the mediating effect of excitement was significant (*effect* = -0.49, *95% CI* = [-0.77, -0.25]), but the mediating effect of relaxation was not (*effect* = -0.04, *95% CI* = [-0.14, 0.05]). Under this condition, assertive messages with virtual anchors increase purchase intention via excitement more than with human anchors. When using non-assertive messages, the mediating effect of excitement was not significant (*effect* = 0.11, *95% CI* = [-0.02, 0.24]), but the mediating effect of relaxation was significant (*effect* = 0.10, *95% CI* = [0.01, 0.20]). Under this condition, non-assertive messages with human anchors increases purchase intention via relaxation more than with virtual anchors. Therefore, H2 and H3 were supported by these results.

We also investigated and ruled out alternative mechanisms using Model 8. The results indicated that the mediating effects of positive mood (*effect* = 0.04, *95% CI* = [-0.04, 0.18]) and perceived enjoyment (*effect* = 0.08, *95% CI* = [-0.02, 0.25]) were not significant.

6.4. Discussion

Study 3 provided further confirmation of the matching effect between anchor type and message assertiveness on purchase intention as well as the mediating mechanisms of excitement and relaxation. The analysis excluded positive mood and perceptive enjoyment as alternative mechanisms. Study 3 replicated Studies 1 and 2 and validated all hypotheses in a controlled setting, enhancing the external validity of the results.

7. General discussion

This study examined the matching effect of anchor type (virtual vs. human) and message assertiveness (assertive vs. non-assertive) on purchase intention, as well as the underlying psychological mechanisms driving this effect. Drawing on the results from two online scenario-based experiments and one laboratory experiment, we found that the interaction between anchor type and message assertiveness positively influenced purchase intention. Relaxation and excitement mediate these effects. Specifically, assertive messages with virtual anchors induced higher excitement levels than with human anchors, leading to increased purchase intention. Conversely, nonassertive messages with human anchors elicited higher relaxation levels than virtual anchors, similarly increasing purchase intention.

Our research advances the understanding of message strategies in live-streaming and describes the emotional differences in consumer responses to assertive messages across anchor types. These findings extend the literature on AI-driven marketing communication, live marketing, and message assertiveness while contributing to the development of language expectancy theory and arousal theory. This study also offers actionable insights for designing effective message strategies for live marketing in live-streaming and similar contexts.

7.1. Theoretical contributions

This study contributes to existing literature in several ways. First, it enriches the growing body of research on AI-driven marketing communications, particularly in the context of virtual anchors. Existing research has predominantly focused on the visual features of virtual anchors (Gao et al., 2023; Stein et al., 2024), whereas more recent studies have explored virtual anchors' use of sensory and arousal language (Wang et al., 2025; Hu and Ma, 2023). However, few studies have addressed the role of assertive message strategies in this context. By demonstrating that assertive messages with virtual anchors and non-assertive messages with human anchors increase purchase intention, our study offers a more comprehensive understanding of how communication strategies affect consumer perceptions and interactions in digital environments.

Second, it augments the existing literature on message assertiveness. Previous studies have explored factors that may influence the effectiveness of assertive messages by considering the characteristics of the product and message recipient (e.g., Kronrod et al., 2012b; Wang, 2022). We agree with Kronrod et al. (2022)'s call for future research to examine other variables related to source characteristics in assertive messaging. This study supplements previous work by focusing on the message source (virtual anchor vs. human anchor). We found that consumers have different psychological and behavioral responses to message assertiveness in live-streaming depending on whether the anchor is human or virtual. This finding expands the scope of message assertiveness research beyond traditional advertising to include live-streaming. In doing so, we offer new insights for the application of message assertiveness research to dynamic, interactive contexts.

Third, our findings extend language expectancy theory by applying it to AI-driven marketing communication with a focus on virtual anchors. Previous language expectancy theory studies primarily examined human-to-human communication (e.g., Li et al., 2023; Ryken et al., 2024). This study identifies how consumer expectations of AI and human anchors influence the matching effect of anchor type and message assertiveness on purchase intention. These results refine language expectancy theory by incorporating the nuances of virtual interactions and broadening its applicability to emerging communication channels.

Finally, this study expands the application of arousal theory to live marketing. It identifies that different anchors using different message assertiveness strategies may stimulate different dimensions of arousal (relaxation and excitement). Previous studies have primarily explored arousal triggered by sensory stimuli, such as color (Gorn et al., 1997) or sound (Biswas et al., 2019). However, our findings reveal that communication strategies can evoke specific types of arousal. Specifically, assertive messages with virtual anchors heighten *consumer excitement* levels, likely because of the novelty and authoritative tone of AI-based communication. In contrast, non-assertive messages with human anchors evoke *relaxation* by aligning with consumers' expectations of a more natural and empathetic interaction. These findings provide a nuanced understanding of how emotional dimensions mediate consumer behavior in live marketing (Liu et al., 2025), suggesting that different combinations of communication strategies and anchor types activate distinct emotional pathways. Moreover, this insight advances our understanding of the antecedents of arousal and their impacts on consumer behavior across diverse platforms and consumer segments.

7.2. Managerial implications

By contrasting the communication styles employed by humans and virtual anchors, this study offers valuable insights for companies engaged in live marketing. Companies should recognize the differences between human and virtual anchors in terms of message assertiveness. Our findings suggest that companies using virtual anchors in live-streaming should employ more assertive messages, such as 'must buy' or 'buy now!'. Conversely, companies using human anchors should

encourage their anchors to use more non-assertive messages, such as 'it's worth it' or 'try it.'

Second, the mediating effects of excitement and relaxation offer actionable insights into how companies can optimize their live-streaming strategies based on anchor type. Virtual anchors generate higher levels of excitement with assertive messages, creating a sense of urgency and energy that motivates impulse purchases. Companies can amplify this effect by incorporating interactive elements such as real-time promotional events, gamification features, or visually dynamic backgrounds to sustain consumer excitement throughout the live-streaming. Conversely, human anchors foster relaxation with non-assertive messages. This aligns with a more conversational and trust-building approach. To capitalize on this, companies should craft a soothing live-streaming atmosphere through warm-toned visuals, calming music, and empathetic storytelling. For example, human anchors can share personal anecdotes about the featured products or use question and answer (Q&A) sessions to engage consumers in a less pressurized and more enjoyable way. Moreover, understanding these mediators equips companies to strategically match their marketing objectives with the right combination of emotional triggers. Excitement-driven strategies may be ideal for high-pressure sales events, such as flash sales. For premium or complex products that require trust and careful consideration, relaxation-driven strategies can help build long-term consumer loyalty.

Finally, this study reveals that consumers have distinct communication expectations based on the type of anchor that delivers the message and that these expectations significantly shape their behavioral responses. While message assertiveness strategies play a critical role, companies can adopt a wider array of tailored communication strategies to effectively address these nuanced preferences. For example, virtual anchors can leverage AI-driven personalization techniques to adjust their language styles and tones according to individual consumer profiles, thereby fostering deeper engagement and higher satisfaction. In addition, businesses may benefit from employing hybrid message strategies that combine assertiveness with other persuasive cues, such as scarcity appeals (e.g., "limited time offer") or social proof (e.g., "best-selling product"), to enhance message impact. These approaches align with the communication expectations associated with each anchor type and increase the message's overall persuasiveness.

7.3. Limitations and future research

First, prior studies have shown that AI characteristics such as appearance (zoonotic vs. robotic) (Poirier et al., 2024), coolness (Gao et al., 2025), and cuteness (Lv et al., 2022), significantly influence consumer behavior. Future research could explore how these features interact with message strategies to shape consumer responses in live-streaming contexts.

Second, this study identified the mediating mechanisms but did not explore potential boundary conditions. For example, consumer characteristics such as hedonic or utilitarian motivations may moderate the matching effect. Hedonic consumers may prioritize enjoyment and interactive experiences, whereas utilitarian consumers may focus on obtaining product information (Gong and Sun, 2025). Additionally, voice could be a factor because users tend to develop more trust in social robots with human voices than in those with synthesized voices (Xu, 2019). Future research should examine these boundary conditions to further contextualize the findings.

Finally, although this study utilized two scenario-based experiments and one laboratory experiment to simulate live-streaming environments, these controlled settings could not fully capture the dynamic and unpredictable nature of real-world live-streaming. Future research could address these limitations by conducting field experiments or partnering with live-streaming platforms to collect real-world data, thereby improving the generalizability of the results, and providing richer insights into consumer behavior in natural settings. In addition, relaxation and excitement were self-reported by the participants, which could have been influenced by social approval bias. Future research could adopt objective physiological measures, such as skin conductance level (SCL) or heart rate (HR), to capture emotional responses more accurately.

CRediT authorship contribution statement

Yanping Gong: Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Rong Huang:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Zhuo Chen:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yongdan Liu:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. **Yuxuan Tan:** Writing – review & editing, Validation, Software, Resources, Investigation.

Ethical statement

All research procedures involving human participants were in accordance with the ethical standards of Ethics Committee of Business School of Central South University and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Acknowledgments

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Appendix A. Demographic information about participants

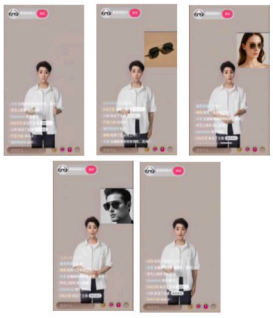
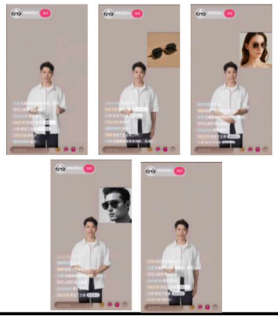
	Study 1	Study 2	Study 3
Sample size	500	431	188
Mean age	30.85	39.59	21.95
Gender			
Female	314	253	136
Male	186	178	52
Education level			
High school and below	26	14	0
Associate degree	46	36	0
Bachelor's degree	354	315	88
Master's degree and above	74	66	100

(*continued on next page*)

(*continued*)

	Study 1	Study 2	Study 3
Monthly living expenses			
Less than 2000 CNY	50	57	41
2001-5000 CNY	81	87	147
5001-10000 CNY	249	193	0
10001-50000 CNY	119	94	0
More than 50,000 CNY	1	0	0
Viewership frequency (in the most recent year)			
1 per month or less	40	80	108
3-5 per month	157	140	53
3-5 times per week	184	161	20
Every day	119	50	7

Appendix B. Stimuli Used in Studies

Study	anchor	Profile
Study 1	Virtual vs. Human	Now I will introduce you to this insulated cup that you must buy/are worth getting. As you can see, this insulated mug has a high-quality, bright, baked enamel body with five colors to choose from. Made from 304 stainless steel, it is glue and odor-free, the liner is food safe, and it costs just 99 (CNY). There's no need to think about it; you've got to get it/You can consider it; it's worth buying to try.
Study 2	Virtual  Human 	Welcome to my live-streaming, and now I'm going to introduce you to the headphones that must buy/are worth getting. This headset is designed with full wrap-around ears, and the shape is very resistant to wear. The material is made from memory foam with zero pressure sensation, which is very comfortable to wear. The battery life is up to 100 hours, with excellent noise cancellation and sound quality. 399 (CNY) price, good value. This is a pair of headphones you will need in your life. You don't have to think about it; you have to get it/You can consider it; it's worth buying to try.
Study 3	Virtual  Human 	Hello everyone, welcome to today's live-streaming! I'm your anchor. The next pair of sunglasses I'm going to share with you are must-buys/worth getting. Let's take a look at these sunglasses first. These sunglasses have UV protection, with oversized lenses, protect your eyes in the sunlight, reduce eyestrain, and make your eyes feel comfortable in any environment. Meanwhile, with their lightweight material, stylish design, and clear lenses, they are a must/could for both beach holidays and everyday travel. Please take a look at the wearing effect of these sunglasses shown in the picture. As you can see, its design is not picky about face shape and can be worn by both men and women. Now we have a special offer: you can enjoy a 10% discount if you place an order for them on the live stream. There's no need to think about it, you've got to get it/You can consider it, it's worth buying to try.

Appendix C. Measurements of main variables

Constructs	Items
Purchase intention	1 I am very interested in purchasing the product introduced by the anchor.
Aaker et al. (2010)	2 I am very likely to purchase the product introduced by the anchor.
	3 I am very satisfied with the product introduced by the anchor.
Consumers' willingness to pay	What is the maximum you would pay for this product?
Schmidt and Bijmolt (2020)	
Relaxation	At this moment, I am feeling:
Gorn et al. (1997)	1 Relaxed
	2 Soothed
	3 Calm
Excitement	At this moment, I am feeling:
Gorn et al. (1997)	1 Excited
	2 Stimulated
Message assertiveness	I think the message introduced by the anchor is:
Baek et al. (2015)	1 Forceful
	2 Intense
	3 Powerful
Perceived price	I think the price of this product is:
Chang and Wildt (1994)	1 = Cheap; 9 = Expensive
Product importance	1 X product is very important to me.
Mittal (1989)	2 For me, X product does not matter (reversed scoring question).
	3 X product is an important part of my life.
Information persuasion	1 I would follow the anchor's recommendation to use this product.
Keller and Block (1996)	2 I am very likely to buy this product.
	3 I would recommend this product to a friend.
	4 I am interested in learning more about this product.
	5 I would discuss this product with a friend.
	6 I wanted to receive additional information about this product.
Perceived competence	I think this anchor is:
Wang et al. (2017)	1 Competent
	2 Intelligent
	3 Capable
	4 Skillful
Perceived warmth	I think this anchor is:
Wang et al. (2017)	1 Warm
	2 Kind
	3 Friendly
	4 Sincere
Positive emotion	1 The degree of happiness you feel.
Kronrod et al. (2012b)	2 The degree of good mood you feel.
Perceptive enjoyment	1 I entertain myself watching this anchor live-streaming.
Soares and Pinho (2014)	2 I like to watch this anchor live-streaming because it is exciting.
	3 Watching this anchor live-streaming makes me happy.
	4 Watching this anchor live-streaming stimulates my curiosity.
	5 Watching this anchor live-streaming stimulates my imagination.
	6 Watching this anchor live-streaming provides relief from a stressful life.

Appendix D. Descriptive statistics and correlations of the variables

Table D.1
Descriptive statistics and correlations of the variables in Study 1

Variables	*M*	*SD*	1	2	3	4	5	6	7
1. Anchor type	0.5	0.5							
2. Message assertiveness	0.5	0.5	0.012						
3. Gender	0.37	0.48	−0.011	−0.038					
4. Age	30.85	9.88	−0.023	−0.063	0.021				
5. Education level	2.95	0.67	−0.042	0.015	−0.087	−0.068			
6. Monthly living expenses	2.88	0.90	−0.006	−0.062	−0.014	0.432***	0.107*		
7. Viewership frequency	2.76	0.90	0.24***	−0.061	0.050	0.137**	0.018	0.158***	
8. Purchase intention	5.07	1.35	−0.063	0.043	0.059	0.240***	0.073	0.294***	0.357***

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. Anchor type: virtual = 0, human = 1; Message assertiveness: assertive = 0, non-assertive = 1; Gender: female = 0, male = 1.

Table D.2
Descriptive statistics and correlations of the variables in Study 2

Variables	*M*	*SD*	1	2	3	4	5	6	7	8	9
1. Anchor type	0.5	0.5									
2. Message assertiveness	0.5	0.5	0.016								
3. Gender	0.41	0.49	−0.029	−0.004							
4. Age	29.59	8.77	−0.022	−0.026	0.070						
5. Education level	3.00	0.61	−0.023	0.031	0.017	0.126***					
6. Monthly living expenses	2.75	0.94	−0.049	−0.091	0.181***	0.491***	0.116*				
7. Viewership frequency	3.41	0.94	0.314***	−0.069	0.035	0.219**	−0.003	0.262***			
8. Excitement	5.00	1.89	−0.070	−0.358***	0.105*	0.139**	−0.055	0.159**	0.134**		
9. Relaxation	5.80	1.82	0.032	0.457***	0.053	0.109*	−0.010	0.007	0.093	−0.400***	
10. Purchase intention	5.20	1.14	−0.026	0.044	0.079	0.186***	−0.014	0.223***	0.318***	0.355***	0.179***

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. Anchor type: virtual = 0, human = 1; Message assertiveness: assertive = 0, non-assertive = 1; Gender: female = 0, male = 1.

Table D.3
Descriptive statistics and correlations of the variables in Study 3

Variables	*M*	*SD*	1	2	3	4	5	6	7	8
1. Anchor type	0.5	0.5								
2. Message assertiveness	0.5	0.5	0.000							
3. Gender	0.28	0.45	0.024	0.024						
4. Age	21.95	2.14	0.000	−0.064	0.027					
5. Monthly living expenses	1.78	0.41	0.090	−0.192**	−0.082	0.059				
6. Viewership frequency	1.61	0.82	0.337***	−0.065	0.180*	0.085	0.092			
7. Excitement	3.54	2.12	−0.207**	−0.463***	0.021	0.125	0.069	0.113		
8. Relaxation	4.77	1.80	0.058	0.401***	−0.126	−0.074	−0.021	0.006	−0.209**	
9. Purchase intention	3.15	1.54	0.051	−0.092	0.132	−0.102	0.041	0.241**	0.502***	0.214**

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. Anchor type: virtual = 0, human = 1; Message assertiveness: assertive = 0, non-assertive = 1; Gender: female = 0, male = 1.

Appendix E. A pre-test of the anchor characteristics in Study 2

Prior to the main study, a pre-test was conducted on the Credamo platform with 200 participants (56% female; M_{age} = 33.06 years) who had experience watching live broadcasts. The participants completed a single-factor (anchor type: virtual vs. human) study in exchange for compensation. Before participating in the pre-test, participants read a brief introduction to online live-streaming and confirmed their experience of watching live-streaming. Subsequently, they viewed the same anchor image data included in the main study and answered questions regarding the perceived authenticity of the live-streaming scenario. This single-item scale was adapted from Cornelis and Peter (2017): "Do you feel that this live-streaming scene is real?" We also used the research method of Zhou et al. (2023) and asked participants to rate the anchor's perceived attractiveness, trustworthiness, and expertise. All items were measured using a 7-point Likert scale ranging from 1 = "not at all" to 7 = "very much." Participants also rated their familiarity with the target anchor (very unfamiliar/familiar, know nothing/know very well) and their general attitude toward the anchor (very negative/positive, very dislike/like) based on their familiarity with the anchor (Zhou et al., 2023). The options were randomized. The results showed no significant difference in perceived authenticity (M_{virtual} = 5.44, SD = 0.99 vs. M_{human} = 5.66, SD = 1.13; $t(198)$ = -1.47, p = 0.144), perceived attractiveness ($M_{virtual}$ = 5.81, SD = 1.12 vs. M_{human} = 5.72, SD = 1.06; $t(198)$ = 0.59, p = 0.559), trustworthiness ($M_{virtual}$ = 5.66, SD = 0.99 vs. M_{human} = 5.54, SD = 1.03; $t(198)$ = 0.80, p = 0.427), expertise ($M_{virtual}$ = 5.75, SD = 1.09 vs. M_{human} = 5.58, SD = 1.10; $t(198)$ = 1.10, p = 0.273), familiarity ($M_{virtual}$ = 3.52, SD = 1.91 vs. M_{human} = 3.70, SD = 1.77; $t(198)$ = -0.67, p = 0.502), and participants' general attitude ($M_{virtual}$ = 5.81, SD = 0.95 vs. M_{human} = 5.77, SD = 0.86; $t(198)$ = 0.31, p = 0.756) between the virtual anchor and the human anchor.

Appendix F. A pre-test of the anchor characteristics in Study 3

Before the main study, a pre-test was conducted in which 60 participants (71.7% female; M_{age} = 21.77 years) who had previous experience watching live-streaming completed a single-factor (anchor type: virtual vs. human) study in the laboratory in exchange for compensation. Participants in the pre-test imagined themselves browsing a shopping platform for a pair of sunglasses, clicking on the detail page, clicking on the live-streaming button in the upper-right corner, and watching the same live-streaming video as in the main study. The same scales used in Study 2 were used to measure perceived situational authenticity, perceived attractiveness, trustworthiness, expertise, familiarity, and general attitude. The results showed no significant difference in perceived authenticity ($M_{virtual}$ = 5.10, SD = 1.86 vs. M_{human} = 5.67, SD = 1.59; $t(58)$ = -1.15, p = 0.255), perceived attractiveness ($M_{virtual}$ = 3.47, SD = 1.46 vs. M_{human} = 4.03, SD = 1.25; $t(58)$ = -1.62, p = 0.111), trustworthiness ($M_{virtual}$ = 3.67, SD = 1.27 vs. M_{human} = 3.90, SD = 1.13; $t(58)$ = -0.75, p = 0.454), expertise ($M_{virtual}$ = 4.00, SD = 1.60 vs. M_{human} = 4.27, SD = 1.31; $t(58)$ = -0.71, p = 0.483), familiarity ($M_{virtual}$ = 2.33, SD = 1.30 vs. M_{human} = 2.47, SD = 1.59; $t(58)$ = -0.36, p = 0.724), and participants' general attitude ($M_{virtual}$ = 4.00, SD = 0.95 vs. M_{human} = 4.13, SD = 1.00; $t(58)$ = -0.53, p = 0.598) between the virtual anchor and the human anchor.

Data availability

Data will be made available on request.

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